Marmara University, Computer Engineering Department

Introduction to Signals and Systems

Online Final Exam

01.07.2020

Solution and Submission Rules

- Textbook, slides and notes are open.
- Internet usage is not allowed; however, computers and cell phones may be used only viewing course materials provided for this course.
- This online exam consists of 4 questions.
- Please write your answers clearly and neatly.
- Show all your work.
- Please draw a box around your final results.

Exam Checklist

☐ I have written my name, surname and student ID on the top of each exam solution page.

☐ I have written the following sentence on the top of the first page with my handwriting and signed it:

"*On my honor, I have neither given nor received any unauthorized assistance on this examination*."

☐ I have solved all the questions with my handwriting on blank A4 pages.

☐ I have scanned all the solution pages to a single PDF file named "myname_surname.pdf".

☐ I have uploaded the PDF file via UES system before the deadline.

1) Consider a discrete LTI system. When the input is -1, 3, 2, 4, the output is 0, -2, 7, 1, 6, -4, 0, 0. If the input is 4, 0, -2, -7, 1, 0, 2 what will be the output?

2) Convolve $tU(t)$ with $-tU(t)$. Evaluate the integrals and plot the result.

3) Compute the Fourier transform of $1 + e^{-3|t|}(2U(t) - 1)$.

4) Consider the signal $f(t)$ whose Fourier transform is

$$F(\omega) = (\omega + 100\pi)\chi_{[-100\pi,0]}(\omega) + (100\pi - \omega)\chi_{[0,100\pi]}(\omega)$$

Draw the signal $f(t)\cos(300\pi t)$ in frequency domain.